Power Meter Analysis & Design

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Introduction

The scope of this report is to analyze the existing power meter included with the Orange Cube Plus and to design a new one from scratch, rated for a total current of at least 250A compared to the 30A maximum rating of the provided module.

Analysis of the included module

No documentation is available, so the PCB was inspected under a microscope to understand how the module operates and communicates with the Cube Orange+.

Three main components can be identified immediately on the PCB:

- A $25\,\mathrm{m}\Omega$ resistor used as a current shunt. The voltage across the shunt follows Ohm's law:

  $$V = IR$$

  At $30\,\mathrm{A}$:

  $$V = 30 \times 0.025 = 0.75\,\mathrm{V}$$

  The power dissipated is:

  $$P = I^2 R = 30^2 \times 0.025 = 22.5\,\mathrm{W}$$

  At $250\,\mathrm{A}$:

  $$V = 250 \times 0.025 = 6.25\,\mathrm{V}, \quad P = 250^2 \times 0.025 = 1562.5\,\mathrm{W} \approx 1.56\,\mathrm{kW}$$

  This would waste significant power and require substantial cooling. A Hall-effect current sensor is preferable, as it adds negligible resistance and dissipates far less power.

- **LT8610**: An LT8610 Synchronous Buck Converter to efficiently step down the high LiPo battery voltage to a stable and accurate $5\,\mathrm{V}$ (fixed version) to power the Cube Orange+ and onboard electronics.

- **INA169**: The INA169 measures the small voltage (0.75V) across the shunt and directly converts it into an output signal using an external resistor that sets the gain. This allows the output voltage to be scaled as needed to 0–$3.3\,\mathrm{V}$ for the ADC.

- An SMD chip marked **8VHAB** - no datasheet found, but probably is an OpAmp Buffer Stage for the ADC signal.

The board connects to the Cube Orange+ through a cable to **POWER1**. Based on [link], it is found that the cable uses a **JST-GH 1.25 mm 6-pin connector** at the power meter end (same type used for other com ports) and a **Molex CLIK-Mate 2.0 mm 6-pin connector** at the Cube Orange+ POWER1 end. The pinout of POWER1 is as follows:

POWER1

Pin	Signal	Volt
1 (red)	VCC	+5V
2 (red)	VCC	+5V
3 (blk)	CURRENT	up to +3.3V,pin 15
4 (blk)	VOLTAGE	up to +3.3V,pin 14
5 (blk)	GND	GND
6 (blk)	GND	GND

Figure 1: POWER1 Port Pinout

both current and voltage sense pins are analog voltages with a range of 0-3.3V.

Design of New Power Meter

Current Meter Choice

The ACS772 (250-CS772ECB300BPFFT) is chosen because it can pass up to 300A, handles calibration internally and outputs an analog voltage which the cube orange + ADC can read.

- **Decoupling capacitor (Cp):** 100 nF mlcc recommended by the datasheet.
- **Low-pass filter (Rf–Cf):** Chosen to capture fast motor current transients while filtering high-frequency noise:

$$R_f = 1\ \text{k}\Omega, \quad C_f = 10\ \text{nF} \quad \Rightarrow \quad f_c = \frac{1}{2\pi R_f C_f} \approx 15.9\ \text{kHz}$$

$$T = \frac{1}{f_c} \approx 63\ \mu\text{s}, \quad \text{which is sufficient to capture signal spikes.}$$

Current Meter Output Translation

According to the datasheet of the ACS772 (250-CS772ECB300BPFFT), the sensor output is **ratiometric around a mid-supply reference**. When powered from V_{CC}:

- **Zero-current output:** approximately $V_{\text{CC}}/2$
- **Sensitivity (300 A version):** approximately 6.66 mV $\times$ $V_{\text{CC}}/5$ per A

The output voltage of the sensor is therefore

$$V_{\text{ACS}} = \frac{V_{\text{CC}}}{2} + \left(6.66\ \text{mV} \cdot \frac{V_{\text{CC}}}{5\ \text{V}}\right) \cdot I$$

For $V_{\text{CC}} = 5$ V and the 300 A measurement range:

$$I = 0\ \text{A} \Rightarrow V_{\text{ACS}} \approx 2.5\ \text{V}$$

$$I = 300\ \text{A} \Rightarrow V_{\text{ACS}} \approx 2.5 + 300 \cdot 6.66\ \text{mV} \approx 4.498\ \text{V}$$

Thus the useful output range of the sensor is approximately

$$2.5\ \text{V} \leq V_{\text{ACS}} \leq 4.498\ \text{V}$$

The CubePilot Orange ADC input range is 0–3.3 V, therefore for maximum accuracy the voltage signal must be translated and scaled to this range. Since only **positive current** is measured, the first step is removing the 2.5 V offset:

$$0\ \text{V} \leq Vcurr \leq 1.998\ \text{V}$$

To utilize the full ADC input range, the signal is then amplified so that 0–1.998 V maps to 0–3.3 V. The required gain is

$$G = \frac{3.3}{1.998} \approx \frac{3.3}{2} = 1.65$$

The final voltage applied to the ADC is therefore

$$0\ \text{V} \leq Vcurr \leq 3.2967\ \text{V}$$

This translation can be implemented using an operational amplifier configured as a **differential amplifier with gain**, using standard resistor values $R_1 = 2\,\text{k}\Omega$ and $R_2 = 3.3\,\text{k}\Omega$ or with **multi-turn** potentiometers in place of R2 for fine tuning. Also any voltage offset can be calibrated through software.

Voltage Meter

For the voltage meter, a simple resistor divider can be used since there is no need for big accuracy (in the decimal range) for the voltage. To scale down the full 6S battery voltage (25.2 V) to the ADC range (0–3.3 V), a division factor of approximately 7.64 is required. Choosing the bottom resistor as $10\,\text{k}\Omega$ and the top resistor as $68\,\text{k}\Omega$ gives:

$$V_{\text{out}} = V_{\text{in}} \cdot \frac{R_{\text{bottom}}}{R_{\text{top}} + R_{\text{bottom}}} = 25.2 \cdot \frac{10}{68 + 10} \approx 3.2308\,\text{V}$$

at maximum battery voltage, which is safely within the ADC limit.

As a result, the usable measurement range for battery voltage is reduced to a window of approximately 0.69 V (2.54 V to 3.23 V). While this represents a loss of theoretical precision, the 12-bit ADC of the Cube Orange+ still provides a resolution of approximately 6.2 mV per bit at the battery level, which is more than sufficient for flight telemetry. The specific voltage and current multipliers can be calibrated within the ArduPilot.

5V LDO Regulator

Since a stable, low noise and low current 5V supply is needed, a linear LDO regulator is chosen. The estimated current draw from the 5 V rail (excluding the LDO's own quiescent current) is calculated as follows:

- **ACS772 Sensor:** According to the datasheet, the typical operating current is 10 mA, with a maximum of 15 mA.
- **MCP6002T Op-Amp:** This dual rail-to-rail op-amp has a maximum quiescent current of $170\,\mu$A per amplifier. For both channels, the total draw is approximately 0.34 mA.
- **V_{CC} Halfer ($V_{CC}/2$):** A $10\,\text{k}\Omega/10\,\text{k}\Omega$ divider is used to create the 2.5 V reference for the differential stage. The current through this branch is:

$$I_{div} = \frac{5\,\text{V}}{10\,\text{k}\Omega + 10\,\text{k}\Omega} = 0.25\,\text{mA}$$

- **Differential Amp - Reference Path:** The non-inverting input of the op-amp is connected to a divider. For $R_1 = 2\,\text{k}\Omega$ and $R_2 = 3.3\,\text{k}\Omega$ connected to the 2.5 V reference:

$$I_{path1} = \frac{2.5\,\text{V}}{R_1 + R_2} = \frac{2.5\,\text{V}}{5.3\,\text{k}\Omega} \approx 0.47\,\text{mA}$$

- **Differential Amp - Feedback Path:** The current through the feedback loop from the output to the inverting input. At max scale ($V_{out} = 3.3\,\mathrm{V}$):

$$I_{path2} \approx \frac{V_{out} - V_{in-}}{R_2} \approx \frac{3.3\,\mathrm{V} - 2.5\,\mathrm{V}}{3.3\,\mathrm{k\Omega}} \approx 0.24\,\mathrm{mA}$$

- **Status LED:** Using a $1\,\mathrm{k\Omega}$ current-limiting resistor for a red SMD LED ($V_f \approx 2\,\mathrm{V}$), the current is:

$$I_{LED} = \frac{5\,\mathrm{V} - 2\,\mathrm{V}}{1000\,\Omega} = 3\,\mathrm{mA}$$

- **Total Estimated Current (I_{total}):** Summing these maximums, the circuit requires approximately $19.30\,\mathrm{mA}$.

While the current is relatively low, the power dissipation (P_D) across the LDO at a maximum 6S battery voltage of $25.2\,\mathrm{V}$ is the primary design constraint:

$$P_D = (V_{in} - V_{out}) \cdot I_{total} = (25.2\,\mathrm{V} - 5\,\mathrm{V}) \cdot 0.0193\,\mathrm{A} \approx 0.39\,\mathrm{W}$$

This remains well within the thermal limits of a SOT-223 package, which typically handles up to $0.5\,\mathrm{W}$ - $1\,\mathrm{W}$.

References

- Allegro MicroSystems ACS772 Current Sensor: `https://gr.mouser.com/ProductDetail/Allegro-MicroSystems/ACS772ECB-300B-PFF-T?qs=HoCaDK9Nz5e9MqYr4%2BWi3Q%3D%3D`
- PCB Headers (RS Online): `https://uk.rs-online.com/web/p/pcb-headers/5468827`
- CubePilot Interface Specifications: `https://docs.cubepilot.org/user-guides/autopilot/the-cube/introduction/interface-specifications`
- Connector Datasheet (Distrelec): `https://media.distrelec.com/Web/Downloads/_t/ds/HX4-06008_eng_tds.pdf`
- ArduPilot Mauch Power Modules Documentation: `https://ardupilot.org/copter/docs/common-mauch-power-modules.html`